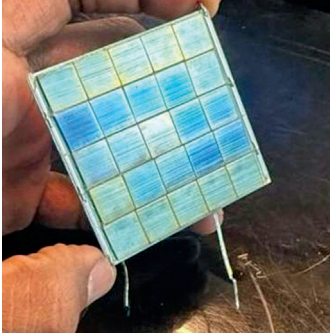


Laser sends 600W of power over 400 metres

Olee Space has demonstrated wireless power transfer by sending 600W of optical power across a 400-metre outdoor line-of-sight link. The system uses an in-house optical



Olee Space demonstrates 600W wireless laser power transmission (Credit: <https://timestech.in>)

receiver that achieved 28% optical-to-electrical conversion efficiency and can scale to 1.5kW with the same panel design. Developed by the Pune-based photonics company, the platform combines beam-shaping optics, thermal management, and position accuracy tracking to maintain alignment and limit heat buildup. The technology could

provide cable-free power for remote infrastructure, 5G base stations, drones, eVTOL platforms, and IoT systems where conventional electrical connections are impractical.

Satellite battery completes first space mission

TakeMe2Space's PowerBank-50 satellite battery has completed its first space mission aboard Skyroot Aerospace's



Aboard Skyroot's Historic Vikram-1, TakeMe2Space's PowerBank-50 becomes the first Made in India, off the shelf satellite battery to earn Flight Heritage on a private Indian Orbital launch

Vikram-1 rocket, gaining flight heritage for future satellite missions. The 50Wh battery integrates lithium-ion cells, a battery management system, heaters, and an aluminium enclosure in a 380g package. During the mission, it powered a payload at about 450km altitude while operating through launch stresses, vacuum, radiation, and temperature variations. The system was developed in India and flew from the Satish Dhawan Space Centre in Sriharikota. Its space-proven design could reduce reliance on imported batteries for CubeSats and microsatellites.

A faster way to build AI camera chips

Researchers from the Chinese University of Hong Kong and the Center for Perceptual and Interactive Intelligence have developed a new way to build ultra-low-power machine vision systems for embedded cameras. Their approach uses a task-agnostic optical neural network that can be fabricated into a chip in about 15 minutes. The research-

ers tested the prototype on several machine vision tasks, including handwritten digit recognition, cell classification, human action recognition, and facial key-point detection. The system achieved classification accuracy between 97% and 99%. By using different materials and nanoimprint replication techniques, future versions could operate across wavelengths ranging from near-ultraviolet to infrared.

New cooling design to protect electronics from overheating

Researchers at the Indian Institute of Technology Madras (IIT Madras) have developed a flat plate pulsating heat pipe design that improves cooling in compact electronic systems.

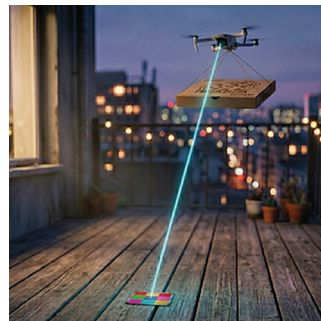


IIT Madras researchers developed cooling design to protect electronics from overheating

The antiparallel layout places the evaporator and condenser on opposite faces, while an O-ring sealing method improves heat transfer by strengthening liquid pulsation. Tests showed 16% lower thermal resistance at higher heat loads than a gasket-based design, while aluminium construction reduced thermal resistance by about 20% compared with copper. Research conducted at IIT Madras could support cooling in smartphones, servers, data centres, defence systems, and electric-vehicle power electronics.

Drone marker enables kilometre identification

Sodyo has introduced RangeMark, a passive visual identification technology that enables drones to identify specific



RangeMark enables drones to identify items from up to one kilometre (Credit: www.prnewswire.com)

targets from up to one kilometre using onboard cameras, without GPS, RF, or network links. The system decodes unique machine-readable markers rather than relying on probabilistic object recognition, enabling deterministic target identification. Developed by Sodyo, the technology uses battery-free printed markers and supports drone operations

at speeds up to 20m/s. Applications include infrastructure inspection, warehouse automation, smart-city asset management, and industrial monitoring. The approach can simplify autonomous operations by reducing dependence on external positioning and communication infrastructure.